IPOD Unification: A Structural Equivalence Framework for Selection-and-Update Phenomena Across Domains

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Abstract

We propose a domain-agnostic formalism for unifying a broad family of "collapse/attention/anchoring/framing"-type phenomena under a single *structural* lens. The core claim is not ontological identity but *structural equivalence*: each phenomenon can be represented as an **IPOD machine** (Input → Process → Output → Data), i.e., a stochastic selection kernel coupled to a state update rule. We formalize IPOD machines as $(\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$ with a Markov kernel $K(\cdot \mid i, d) \in \Delta(\mathcal{O})$ and an update map $U : \mathcal{D} \times \mathcal{I} \times \mathcal{O} \to \mathcal{D}$. We then define a notion of structural equivalence via commuting encoders/decoders between machines, prove representation results (by explicit construction) for quantum measurement instruments, transformer next-token generation, anchoring as score distortion, and framing as meta-constraints shaping the score field. We also show that for finite outputs, any strictly positive kernel admits a softmax (log-probability) coordinate representation, clarifying the role of softmax as a convenient parameterization rather than a universal physical law.

Keywords. Structural equivalence; selection kernels; IPOD; softmax; quantum instruments; attention; anchoring; framing.

1 Introduction

Across physics, cognition, AI, and social sciences, one repeatedly encounters systems that begin with a space of possibilities and end with a single expressed, reported, or stabilized outcome. Prominent examples include quantum measurement outcomes (often informally called "collapse"), transformer attention-driven token selection, anchoring effects in judgment, and framing effects in discourse.

A common failure mode in cross-domain synthesis is *frame drift*: treating distinct domain concepts as identical in meaning or ontology. This manuscript adopts a stricter objective: *prove structural similarity* by representing each phenomenon within the same abstract machine schema, without claiming that the underlying ontologies or mechanisms are the same.

Contributions.

- We define an **IPOD machine** as a minimal selection-and-update system:

$$O_t \sim K(\cdot \mid I_t, D_t), \qquad D_{t+1} = U(D_t, I_t, O_t).$$

- We define **structural equivalence** of IPOD machines via commuting maps between inputs/outputs/states that preserve the induced output distributions and state transitions.
- We prove an **IPOD representation theorem** by construction for: (i) quantum instruments (POVM/Kraus update), (ii) transformer next-token generation, (iii) anchoring as a

distortion of a latent score field, (iv) framing as a meta-process constraining the score field and update dynamics.

- We prove that for finite outputs with full support, *any* kernel can be written as a softmax of a score function, clarifying what is (and is not) implied by "softmax unification".

2 Preliminaries and Notation

Let $\Delta(\mathcal{O})$ denote the set of probability distributions over a measurable output space $\mathcal{O}$ (for finite $\mathcal{O}$ this is the simplex). A (Markov) kernel $K : \mathcal{I} \times \mathcal{D} \to \Delta(\mathcal{O})$ assigns to each (i, d) a distribution $K(\cdot \mid i, d)$.

3 IPOD Machines

Definition 1 (IPOD machine)**.** An *IPOD machine* is a tuple

$$\mathcal{M} = (\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$$

where:

- $\mathcal{I}$ is an input space, $\mathcal{O}$ an output space, $\mathcal{D}$ a data/state space,
- $K : \mathcal{I} \times \mathcal{D} \to \Delta(\mathcal{O})$ is a selection kernel,
- $U : \mathcal{D} \times \mathcal{I} \times \mathcal{O} \to \mathcal{D}$ is a state update map.

The machine evolves as

$$O_t \sim K(\cdot \mid I_t, D_t), \qquad D_{t+1} = U(D_t, I_t, O_t). \tag{1}$$

Remark 1 (Interpretation discipline)**.** The symbols $(\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$ are *structural* slots. A domain-specific interpretation assigns meanings to these slots (e.g. $\mathcal{D}$ as a density matrix in quantum theory, or as cultural memory in sociology). Structural unification concerns the existence of such an assignment, not identity of the underlying ontology.

4 Structural Equivalence of IPOD Machines

Definition 2 (Structural equivalence)**.** Let $\mathcal{M} = (\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$ and $\mathcal{M}' = (\mathcal{I}', \mathcal{O}', \mathcal{D}', K', U')$ be IPOD machines. A *structure-preserving morphism* $\Phi : \mathcal{M} \to \mathcal{M}'$ is a triple of maps

$$\phi_I : \mathcal{I} \to \mathcal{I}', \quad \phi_O : \mathcal{O} \to \mathcal{O}', \quad \phi_D : \mathcal{D} \to \mathcal{D}'$$

such that for all $(i, d) \in \mathcal{I} \times \mathcal{D}$:

(i) (*Kernel commutation / pushforward*) the distribution of $\phi_O(O)$ under $K(\cdot \mid i, d)$ equals $K'(\cdot \mid \phi_I(i), \phi_D(d))$:

$$(\phi_O)_\# K(\cdot \mid i, d) = K'(\cdot \mid \phi_I(i), \phi_D(d)). \tag{2}$$

(ii) (*Update commutation*)

$$\phi_D(U(d, i, o)) = U'(\phi_D(d), \phi_I(i), \phi_O(o)), \quad \forall o \in \mathcal{O}. \tag{3}$$

If there exist morphisms $\Phi : \mathcal{M} \to \mathcal{M}'$ and $\Psi : \mathcal{M}' \to \mathcal{M}$ that are inverses on the represented behavior, we call $\mathcal{M}$ and $\mathcal{M}'$ *structurally equivalent.*

Remark 2. This definition encodes what "same structure" means operationally: selection distributions and state transitions commute under representation maps. It is the formal counterpart of "same IPOD schema".

5 Softmax as a Coordinate Representation (Not an Ontology)

Lemma 1 (Softmax representation of any full-support finite kernel)**.** *Assume $\mathcal{O}$ is finite and $K(o \mid i, d) > 0$ for all $o \in \mathcal{O}$. Define the score field*

$$A(o \mid i, d) = \log K(o \mid i, d).$$

Then for all (i, d),

$$K(o \mid i, d) = \frac{e^{A(o|i,d)}}{\sum_{o' \in \mathcal{O}} e^{A(o'|i,d)}} = \mathrm{Softmax}(A(\cdot \mid i, d))(o). \tag{4}$$

Proof. Substitute $A = \log K$ into the softmax. The numerator becomes $K(o \mid i, d)$ and the denominator becomes $\sum_{o'} K(o' \mid i, d) = 1$. □

Remark 3 (Why this matters)**.** Claims of the form "everything is softmax" can be misleading. Lemma 1 shows softmax is a *parameterization* of any finite full-support kernel in log-probability coordinates. Thus, the true invariant is the kernel K (selection structure), not the particular coordinate form used to represent K.

6 Representation Theorem: Domains as IPOD Machines

Theorem 1 (IPOD representation theorem (structural unification))**.** *Each of the following can be represented as an IPOD machine $\mathcal{M} = (\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$ in the sense of* **??***, with domain-appropriate instantiations:*

(a) quantum measurement as a quantum instrument;
(b) transformer next-token generation;
(c) anchoring as score-field distortion in a choice/estimation kernel;
(d) framing as a meta-constraint shaping the score-field and update.

Proof by explicit construction. We construct $(\mathcal{I}, \mathcal{O}, \mathcal{D}, K, U)$ for each domain.

(a) Quantum measurement (instrument form). Let $\mathcal{D}$ be the set of density operators ρ on a Hilbert space. Let $\mathcal{I}$ be the set of measurement setups M represented by a POVM $\{E_k\}_{k \in \mathcal{O}}$ with outcomes $\mathcal{O} = \{k\}$. Define the kernel

$$K(k \mid M, \rho) = \mathrm{Tr}(E_k \, \rho). \tag{5}$$

Define the update (Lüders/Kraus form; one standard choice)

$$U(\rho, M, k) = \rho_k = \frac{\sqrt{E_k} \, \rho \, \sqrt{E_k}}{\mathrm{Tr}(E_k \rho)}. \tag{6}$$

Then (1) holds by definition: an outcome is sampled from K, and the post-measurement state is updated by U. This is precisely an IPOD machine with $I = M$, $D = \rho$, $O = k$.

(b) Transformer next-token generation. Let $\mathcal{I}$ be the set of token prefixes $x_{1:t}$. Let $\mathcal{D}$ encode model parameters and runtime state (e.g. KV-cache). Let $\mathcal{O}$ be the vocabulary. Define logits $\ell(\cdot \mid x_{1:t}, D_t)$ produced by the transformer forward pass. Set

$$K(x_{t+1} \mid x_{1:t}, D_t) = \mathrm{Softmax}(\ell(\cdot \mid x_{1:t}, D_t))(x_{t+1}), \tag{7}$$

and define U as the deterministic update that appends the token and updates cache/state. Thus (1) holds.

(c) Anchoring (bias as score distortion). Let $\mathcal{O}$ be a discretized set of estimates (or choices) y. Let $\mathcal{I}$ include a question q and anchor value a. Let $\mathcal{D}$ be an agent memory/prior state. Let $A_0(y \mid q, d)$ be a baseline score field absent anchoring, and let $g(y, a)$ encode anchor-induced pull; $\lambda \geq 0$ is anchor strength. Define

$$A(y \mid (q, a), d) = A_0(y \mid q, d) + \lambda\, g(y, a), \tag{8}$$

and choose any kernel consistent with this score (e.g. softmax for finite $\mathcal{O}$):

$$K(y \mid (q, a), d) = \mathrm{Softmax}(\beta A(\cdot \mid (q, a), d))(y). \tag{9}$$

Let U update memory with (q, a, y) (priming/consolidation). Then the anchoring process is an IPOD machine. Importantly, anchoring is not "the same mechanism" as transformer attention; it is an instantiation where the anchor enters by deforming A.

(d) Framing (meta-constraint shaping the score field). Let $\mathcal{O}$ be a set of frames f applicable to an event $e \in \mathcal{I}$. Let $\mathcal{D}$ encode cultural/institutional state. Let the score field

$$A(f \mid e, d) = \mathrm{fit}(f, e, d) + \mathrm{influence}(f, d) - \mathrm{cost}(f, d)$$

aggregate explanatory fit and social power components (domain-specific). Define a selection kernel (again, softmax is one option when $\mathcal{O}$ is finite):

$$K(f \mid e, d) = \mathrm{Softmax}(\beta A(\cdot \mid e, d))(f).$$

Let U update d by institutionalizing the selected frame f given event e. Then framing is an IPOD machine: a selection kernel on frames plus a social-memory update.

Each construction yields an IPOD machine as claimed. □

Remark 4 (What the theorem does *not* claim)**.** Theorem 1 proves *representability* in a shared schema (selection kernel + state update), i.e. structural unification. It does not claim that quantum outcomes are literally generated by the same algorithm as transformer attention, nor does it derive the Born rule from softmax.

7 Unifying "Collapse = Attention" After Audit: The Correct Statement

With structural discipline, the unified statement is:

> *Quantum measurement outcomes, transformer token generation, anchoring in estimation, and framing in discourse are structurally unified as IPOD machines: each consists of (i) an input-conditioned stochastic selection kernel over a possibility set, and (ii) a state/data update rule that feeds forward into subsequent selections.*

Softmax may appear as: (i) a genuine model choice (e.g. logit / entropy-regularized selection), or (ii) a coordinate representation of any finite full-support kernel (Lemma 1). Thus, "softmax everywhere" is best read as either a modeling hypothesis or a re-parameterization, while the robust invariant is the IPOD kernel-update structure.

8 Discussion: Where the Real Cross-Domain "Connections" Live

Connection 1: Selection structure. All domains share the map $(I, D) \mapsto K(\cdot \mid I, D)$ producing a distribution on possibilities.

Connection 2: Feedback / path dependence. The update $D_{t+1} = U(D_t, I_t, O_t)$ encodes priming, reinforcement, institutionalization, KV-cache accumulation, or post-measurement state update.

Connection 3: Regimes (soft vs hard). "Soft" vs "hard" selection is captured by different kernels (or temperature limits), but both sit inside the same IPOD schema.

9 Limitations and Falsifiable Implications (Structural, Not Ontological)

Limitations. Structural unification does not settle interpretation debates (e.g. in quantum foundations), nor does it by itself predict domain-specific constants or mechanistic micro-details.

Empirical implications. The framework suggests a measurable decomposition in each domain: (i) infer a kernel K (choice probabilities, token probabilities, outcome statistics), (ii) infer an update U (memory/culture/state transition), (iii) test whether repeated behavior is consistent with an IPOD loop (e.g. Markovianity in an augmented state, or systematic score-field deformation).

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